OKLAHOMA SCHOOL TESTING PROGRAM

TEST BLUEPRINT AND ITEM SPECIFICATIONS

GRADE 4 MATHEMATICS

OKLAHOMA Education

1 Which equation is true?

A $8{,}000 = 80{,}000 \times 10$

B $8{,}000 = 80{,}000 \times 100$

C $80{,}000 = 8{,}000 \times 10$

D $80{,}000 = 8{,}000 \times 100$

Standard: 4.N.1.3 Applying knowledge of place value, use mental strategies (no written computations) to multiply or divide a number by 10, 100, and 1,000.

Depth-of-Knowledge: 1

This item is DOK 1 because it requires the student to perform a simple procedure, multiplying whole numbers by 10 and 100.

Distractor Rationale:

A. The student made a place value error.
B. The student made a place value error.
C. Correct. The student demonstrated an ability to apply knowledge of place value to multiply a number by 10 or 100.
D. The student made a place value error.

2 Which number is **greater** than 204,320?

A 201,450

B 205,119

C 204,307

D 200,999

Standard: 4.N.1.4 Use place value to compare and order whole numbers up to 1,000,000, using comparative language, numbers, and symbols.

Depth-of-Knowledge: 2

This item is DOK 2 because it requires the student to compare numbers.

Distractor Rationale:

A. The student compared 320 to 450 only.
B. Correct. The student demonstrated an ability to compare whole numbers.
C. The student confused 204,307 and 204,370.
D. The student compared 320 to 999 only.

3 For a school activity, 96 students will work in groups. Each group will have 12 students.

How many groups will there be?

A 6 groups

B 7 groups

C 8 groups

D 9 groups

Standard: 4.N.2.1 Demonstrate fluency with multiplication and division facts with factors up to 12.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine that this is a division scenario and then to compute.

Distractor Rationale:

A. The student made a fact error.
B. The student made a fact error.
C. Correct. The student demonstrated an ability to fluently divide with factors up to 12.
D. The student made a fact error.

4 Students from a school are going to a theme park. The cost for each student ticket is \$27. The school will buy 32 student tickets.

What is the total cost of these tickets?

A \$288

B \$654

C \$801

D \$864

Standard: 4.N.2.2 Multiply 3-digit by 1-digit and 2-digit by 2-digit whole numbers, using various strategies, including but not limited to standard algorithms.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine that this is a multiplication scenario and then to compute the total.

Distractor Rationale:

A. The student computed using the standard algorithm and computed 2 × 32 instead of 20 × 32 in the second row.
B. The student did not know how to multiply two two-digit numbers.
C. The student did not know how to multiply two two-digit numbers.
D. Correct. The student demonstrated an ability to multiply two two-digit numbers.

5 Gretta planted 24 rows of carrots. Each row had 16 carrots in it.

Which is **closest** to the total number of carrots Gretta planted?

A 200 carrots

B 300 carrots

C 400 carrots

D 600 carrots

Standard: 4.N.2.3 Estimate products of 3-digit by 1-digit and 2-digit by 2-digit whole number factors using a variety of strategies (e.g., rounding, front end estimation, adjusting, compatible numbers) to assess the reasonableness of results. Explore larger numbers using technology to investigate patterns.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to multiply and then make a decision about which answer is closest.

Distractor Rationale:

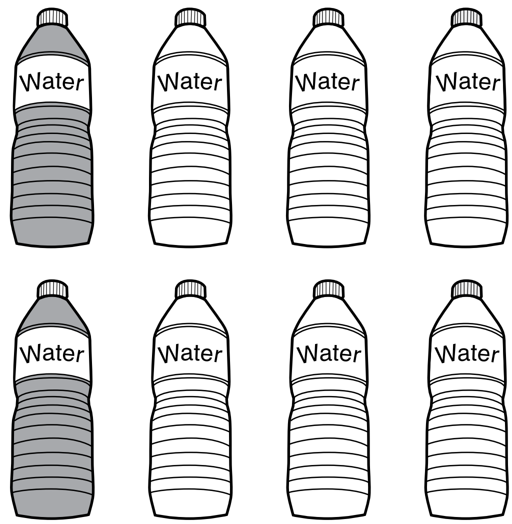
A. The student incorrectly rounded 16 to 10.

B. The student found the product and then only focused on the hundreds place being 3.

C. Correct. The student demonstrated an ability to estimate the product of a 2-digit by 2-digit multiplication problem using rounding.

D. The student incorrectly rounded 24 to 30.

6 The picture shows 8 water bottles.



Which pair of equivalent fractions shows the part of the group of water bottles that is shaded?

A $\frac{6}{8} = \frac{3}{4}$

B $\frac{2}{8} = \frac{1}{3}$

C $\frac{6}{8} = \frac{2}{3}$

D $\frac{2}{8} = \frac{1}{4}$

Standard: 4.N.3.1 Represent and rename equivalent fractions using fraction models (e.g., parts of a set, area models, fraction strips, number lines).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to interpret the model, turn the model into a fraction, and then represent that fraction as an equivalent fraction.

Distractor Rationale:

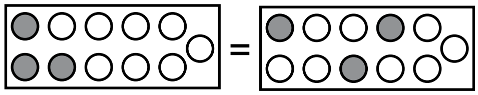
A. The student identified the part of the group that is not shaded.

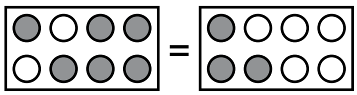
B. The student thought $\frac{2}{8}$ was equivalent to $\frac{1}{3}$.

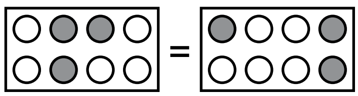
C. The student identified the part of the group that is not shaded and thought $\frac{6}{8}$ was equivalent to $\frac{2}{3}$.

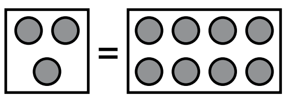
D. Correct. The student demonstrated an ability to identify a fraction shown in a model and rename that fraction as an equivalent fraction.

7 **Which diagram <u>best</u> represents $\frac{3}{8} = \frac{3}{8}$?**

A 

B 

C 

D 

Standard: 4.N.3.3 Use models to order and compare whole numbers and fractions less than and greater than one, using comparative language and symbols.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to identify a model that represents an equation with fractions. The models for the same fraction are not identical, requiring the student to think about different ways to model the same number.

Distractor Rationale:

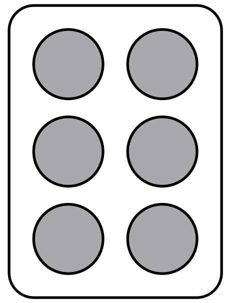
A. The student thought $\frac{3}{8}$ means 3 shaded and 8 not shaded.

B. The student saw $\frac{3}{8}$ on the right side of the equation.

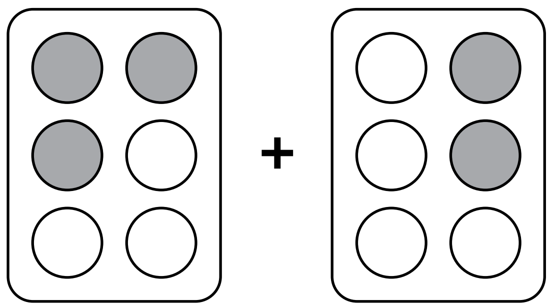
C. Correct. The student demonstrated an ability to use models to compare fractions.

D. The student saw 3 on the left and 8 on the right.

8 This is a full package of muffins, which represents one whole.



These two packages of muffins were on the counter.



What fraction of a full package of muffins were on the counter?

A $\frac{5}{12}$

B $\frac{1}{12}$

C $\frac{5}{6}$

D $\frac{1}{6}$

Standard: 4.N.3.5 Use models to add and subtract fractions with like denominators.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure.

Distractor Rationale:

A. The student added the numerators and the denominators.

B. The student subtracted instead of added the numerators and added the denominators.

C. Correct. The student demonstrated an ability to add fractions with like denominators using a model.

D. The student subtracted instead of added.

9 **Gigi ran one mile for her gym class. It took her 14.79 minutes to run the mile. Which shows 14.79 in word form?**

A fourteen and seventy-nine ones

B fourteen and seventy-nine tenths

C fourteen and seventy-nine hundredths

D fourteen and seventy-nine thousandths

Standard: 4.N.3.7 Read and write decimals in standard, word, and expanded form up to at least the hundredths place in a variety of contexts, including money.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, representing a number using words.

Distractor Rationale:

A. The student confused ones and hundredths.

B. The student confused tenths and hundredths.

C. Correct. The student demonstrated an ability to read and write a decimal up to the hundredths place.

D. The student confused thousandths and hundredths.

10 The table shows the number of inches of rainfall each month.

Rainfall

Month	Rainfall (inches)
March	2.53
April	2.6
May	2.08

Which list shows these rainfall amounts in order from **least** to **greatest**?

A 2.6, 2.53, 2.08

B 2.08, 2.53, 2.6

C 2.53, 2.6, 2.08

D 2.08, 2.6, 2.53

Standard: 4.N.3.8 Compare and order decimals and whole numbers using place value and various models including but not limited to grids, number lines, and base 10 blocks.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to compare and organize the data presented.

Distractor Rationale:

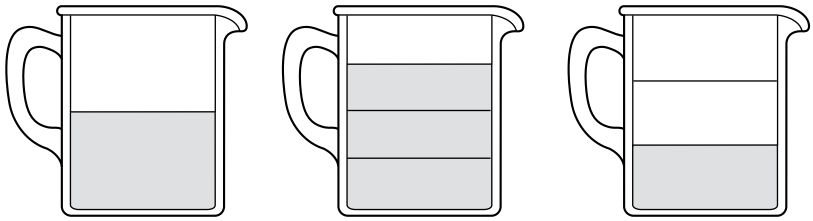
A. The student ordered from greatest to least.

B. Correct. The student demonstrated an ability to order decimals.

C. The student used the same order as the table.

D. The student thought 2.53 was larger than 2.6 because it had more digits.

11 **The three cups shown are the same size. Each cup has a different amount of juice.**



$\frac{1}{2}$ $\frac{3}{4}$ $\frac{1}{3}$

Which list shows the amounts in order from greatest to least?

A $\frac{3}{4}, \frac{1}{3}, \frac{1}{2}$

B $\frac{1}{2}, \frac{1}{3}, \frac{3}{4}$

C $\frac{3}{4}, \frac{1}{2}, \frac{1}{3}$

D $\frac{1}{2}, \frac{3}{4}, \frac{1}{3}$

Standard: 4.N.3.9 Compare and order benchmark fractions (0, $\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$, 1) and decimals (0, 0.25, 0.50, 0.75, 1.00) in a variety of representations.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to order fractions with unlike denominators from greatest to least.

Distractor Rationale:

A. The student focused only on the denominators.

B. The student reversed $\frac{1}{2}$ and $\frac{1}{3}$.

C. Correct. The student demonstrated an ability to compare benchmark fractions.

D. The student did not know how large $\frac{1}{3}$ is.

12 **Mr. Wilson charges a customer $7 for a new toy. The customer pays Mr. Wilson with a $20 bill. How much change does Mr. Wilson owe the customer?**

A $3

B $5

C $10

D $13

Standard: 4.N.4.2 Given a total cost (dollars and coins up to twenty dollars) and amount paid (dollars and coins up to twenty dollars), find the change required in a variety of ways.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the change after a whole dollar payment.

Distractor Rationale:

A. The student only gave change for a $10 bill or made a regrouping error when subtracting.

B. The student computed 7−2.

C. The student ignored the one dollar bills.

D. Correct. The student demonstrated an ability to find the change required given an amount paid (whole dollars up to twenty).

13 **Jane bought a pencil for 21¢. She used a quarter to pay for the pencil. How much change should Jane get back?**

A 1¢

B 3¢

C 4¢

D 5¢

Standard: 4.N.4.2 Given a total cost (dollars and coins up to twenty dollars) and amount paid (dollars and coins up to twenty dollars), find the change required in a variety of ways.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to know the value of a quarter and apply this to paying for an item and figuring out the change.

Distractor Rationale:

A. The student thought a quarter was worth 20¢ and then computed 21−20 instead of 20−21.

B. Balance distractor

C. Correct. The student demonstrated an ability to find the change required given an amount paid (coins).

D. The student used 20¢ for the cost of the pencil.

14 **A function machine used the rule multiply by 6. Which table could represent the numbers going in and coming out of this function machine?**

A

In	Out
2	8
3	9
6	12
8	14

B

In	Out
2	12
5	30
8	48
9	54

C

In	Out
1	6
3	18
5	30
7	48

D

In	Out
2	12
3	18
6	24
8	30

Standard: 4.A.1.1 Create an input/output chart or table to represent or extend a numerical pattern.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to analyze each input/output table to identify the one that matches a given rule.

Distractor Rationale:

A. The student confused multiply by 6 and add 6.

B. Correct. The student demonstrated an ability to create and input/output table to represent a numerical pattern.

C. The student saw that this worked for the first 3 inputs.

D. The student saw that this worked for the first 2 inputs.

15 **The table shows the cost of different numbers of tickets to a baseball game.**

Baseball Tickets

Number of Tickets (t)	Cost ($)
2	16
3	24
4	32
5	40

Which rule can be used to find the cost, in dollars, of *t* tickets?

A $t \bullet 8$

B $t \div 12$

C $t + 14$

D $t - 35$

Standard: 4.A.1.2 Describe the single operation rule for a pattern from an input/output table or function machine involving any operation of a whole number.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine and then describe the rule shown in an input/output table.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to describe the single operation rule for a pattern presented in a table.**
B. Balance distractor
C. The student saw that this rule worked for 2 tickets.
D. The student thought this worked for 5 tickets, but the relationship is reversed.

16 **The first four figures of a geometric pattern of 1-unit squares are shown.**

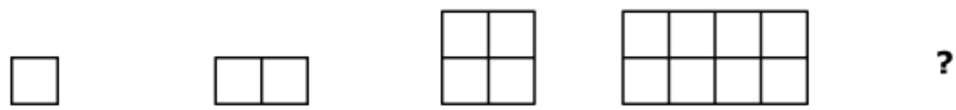


Figure 1 Figure 2 Figure 3 Figure 4 Figure 5

When the pattern continues, what rule can be used to find the number of 1-unit squares in Figure 5?

A Add 2.

B Add 4.

C Multiply by 2.

D Multiply by 4.

Standard: 4.A.1.3 Construct models to show growth patterns involving geometric shapes and define the single operation rule of the pattern.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine and then describe the rule shown in a geometric pattern.

Distractor Rationale:

A. The student saw that this rule worked from Figure 2 to Figure 3.

B. The student saw that this rule worked from Figure 3 to Figure 4.

C. Correct. The student demonstrated an ability to extend a growth pattern involving geometric shapes and define the single operation rule of the pattern.

D. Balance distractor

17 **Coach Ted bought 36 banners. He bought an equal number of blue banners and gold banners. The number of banners of each color, *n*, can be found using this equation.**

$$\boxed{2 \times n = 36}$$

How many banners of each color did Coach Ted buy?

A 18 banners

B 34 banners

C 38 banners

D 72 banners

Standard: 4.A.2.1 Use the relationships between multiplication and division with the properties of multiplication to solve problems and find values for variables that make number sentences true.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to develop a strategy for finding the missing factor in a number sentence.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to use the relationship between multiplication and division to solve a problem and find an unknown represented by a letter.**

B. The student computed 36 − 2.

C. The student computed 36 + 2.

D. The student computed 36 × 2.

18

Match each equation on the left to the correct value for *n* on the right. Each equation on the left matches to one value of *n* on the right. Click one box on the left and then click its match on the right. To remove a line, hold the pointer over the line until it turns red, and then click it.

$5 \times n = 40$	$n = 3$
$7 \times n = 35$	$n = 5$
$12 \times n = 36$	$n = 8$

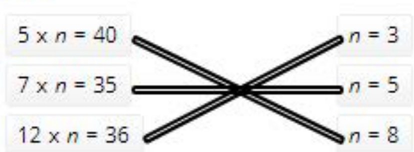
Standard: 4.A.2.1 Use the relationships between multiplication and division with the properties of multiplication to solve problems and find values for variables that make number sentences true.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to develop a strategy for finding the missing factor number sentences.

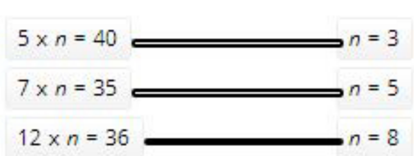
Distractor Rationale:

Correct



Incorrect

The student thought that both factors should be the same for the first equation.



The student thought that the smallest factor should be matched with the other smallest factor, the middle one with the middle, and the largest factor with the other largest factor.

19 **Marcia is making chocolate chip cookies. She needs to use a total of 64 ounces of chocolate chips. She already has 16 ounces of chocolate chips. The equation can be used to find the number of ounces of chocolate chips, *c*, Marcia still needs to use.**

$$\boxed{16 + c = 64}$$

How many ounces of chocolate chips does Marcia still need to use?

A 48 ounces

B 52 ounces

C 58 ounces

D 80 ounces

Standard: 4.A.2.2 Solve for unknowns in problems by solving open sentences (equations) and other problems involving addition, subtraction, multiplication, or division with whole numbers. Use real-world situations to represent number sentences and vice versa.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to develop a strategy for finding the unknown in a number sentence.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to solve for an unknown by solving an equation involving addition with whole numbers.**
B. The student made a computational error.
C. The student made a computational error.
D. The student computed 64 + 16.

20 A student drew a four-sided polygon. Each side of the polygon was the same length and had no right angles.

Which is the **best** name for the polygon this student drew?

A trapezoid

B square

C rhombus

D rectangle

Standard: 4.GM.1.2 Describe, classify, and construct quadrilaterals, including squares, rectangles, trapezoids, rhombuses, parallelograms, and kites. Recognize quadrilaterals in various models.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use an understanding of the characteristics of polygons to determine the best name for the polygon described.

Distractor Rationale:

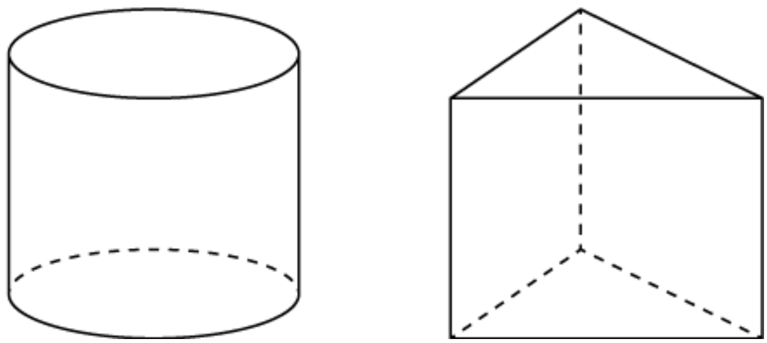
A. The student did not understand the attributes of a trapezoid.

B. The student focused on the four congruent sides and ignored the part about the angles.

C. Correct. The student demonstrated an understanding of the attributes of quadrilaterals.

D. The student did not understand the attributes of a rectangle.

21 **A cylinder and triangular prism are shown.**



Which statement about these figures is true?

A The cylinder and prism each have the same number of faces.

B The cylinder and prism each have the same number of edges.

C The cylinder and prism each have 6 vertices.

D The cylinder and prism each have 2 bases.

Standard: 4.GM.1.3 Given two three-dimensional shapes, identify each shape. Compare and contrast their similarities and differences based on their attributes.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to compare two different three-dimensional shapes.

Distractor Rationale:

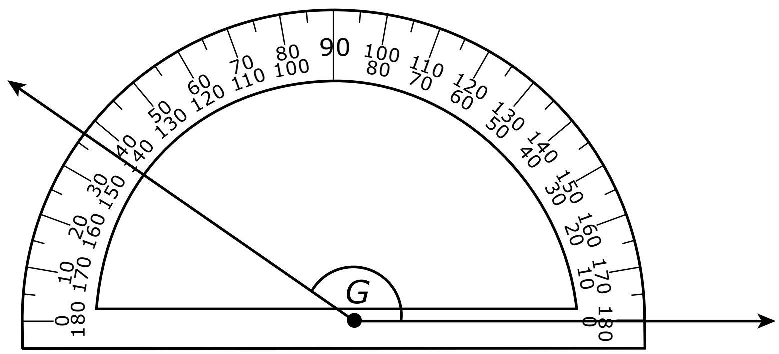
A. The student confused faces and bases.

B. The student confused edges and bases.

C. The student confused vertices and bases.

D. Correct. The student demonstrated an ability to identify similarities and differences given two three-dimensional shapes.

22 **Which is <u>closest</u> to the measure of $\angle G$?**



A 37°

B 43°

C 143°

D 157°

Standard: 4.GM.2.1 Measure angles in geometric figures and real-world objects with a protractor or angle ruler.

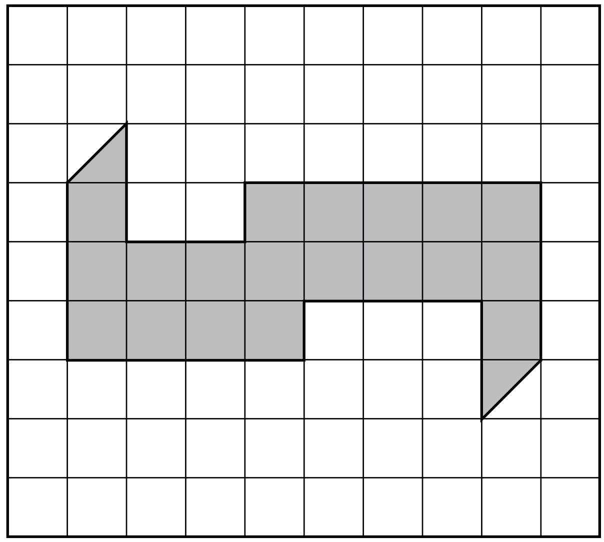
Depth-of-Knowledge: 2

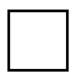
This item is a DOK 2 because it requires the student to measure using a protractor when the measurement is not exactly on a labeled line.

Distractor Rationale:

A. The student used the wrong set of numbers on the protractor.
B. The student used the wrong set of numbers on the protractor and thought the line is 3 degrees past 40 degrees.
C. Correct. The student demonstrated an ability to measure angles with a protractor.
D. The student misread the protractor and thought the line is 7 degrees past 150 degrees.

23 **The shape of a parking lot at a shopping mall is represented by the shaded part of this grid.**



 = 1 square unit

What is the area, in square units, of the parking lot?

A 19 square units

B 20 square units

C 21 square units

D 28 square units

Standard: 4.GM.2.2 Find the area of polygons by determining if they can be decomposed into rectangles.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to come up with a strategy for finding the area of a figure on a grid that can be decomposed into rectangles.

Distractor Rationale:

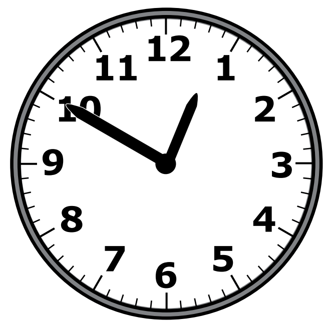
A. The student subtracted the area of the triangles.

B. Correct. The student demonstrated an ability to find the area of polygons that can be decomposed into rectangles.

C. The student thought each triangle was worth 1 square unit.

D. The student found approximate perimeter instead of area.

24 **Carmen started eating her snack at the time shown on the clock.**



It took Carmen 15 minutes to eat her snack. At what time did Carmen finish eating her snack?

A 1:05

B 2:05

C 10:20

D 12:35

Standard: 4.GM.3.1 Determine elapsed time.

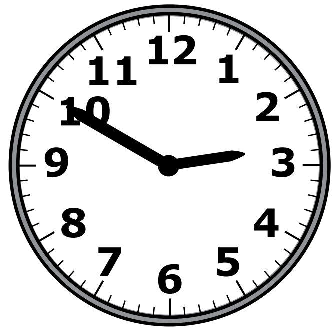
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to first decide how to approach the problem and then determine elapsed time.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to determine elapsed time.**
B. The student thought the clock showed 1:50 because the hour hand is closer to the 1 than the 12.
C. The student thought the clock showed 10:05.
D. The student subtracted 15 minutes from the time shown.

25 **A movie begins at 2:50 P.M. as shown on this clock.**



The movie ends at 4:30 P.M. How long is the movie?

A 1 hour 30 minutes

B 1 hour 35 minutes

C 1 hour 40 minutes

D 1 hour 45 minutes

Standard: 4.GM.3.1 Determine elapsed time.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to first decide how to approach the problem and then determine elapsed time.

Distractor Rationale:

A. The student chose because the ending time has 30 minutes in it.

B. Balance distractor

C. Correct. The student demonstrated an ability to determine elapsed time.

D. Balance distractor

26 **A coach timed all of the students in a class to see how long it took them to finish a race. Sam finished the race in 180 seconds.**

How many minutes did it take Sam to finish the race?

A 1 minute

B 2 minutes

C 3 minutes

D 4 minutes

Standard: 4.GM.3.2 Convert one measure of time to another including seconds to minutes, minutes to hours, hours to days, and vice versa, using various models.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate from seconds to minutes.

Distractor Rationale:

A. The student thought there are 180 seconds in 1 minute.

B. The student thought there are 90 seconds in 1 minute.

C. Correct. The student demonstrated an ability to convert between seconds and minutes.

D. The student thought there are 45 seconds in 1 minute.

27 **Joy and Fran each have some toy horses.**

Joy's Horses

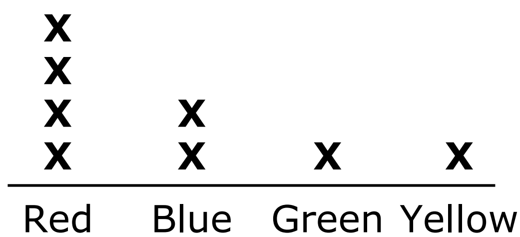
Color	Number of Horses
Red	1
Blue	1
Green	2
Yellow	1

Fran's Horses

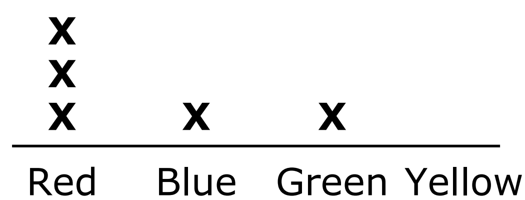
Color	Number of Horses
Red	3
Blue	1
Green	1
Yellow	0

Which line plot shows how many horses of each color the girls have all together?

A



B



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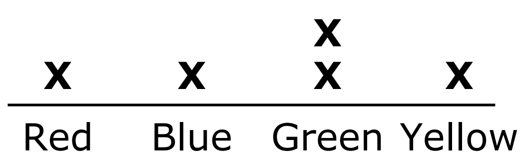
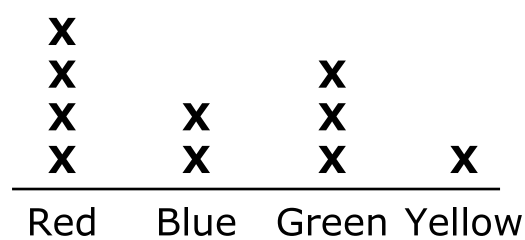


D



Standard: 4.D.1.1 Create and organize data on a frequency table or line plot marked with whole numbers and fractions using appropriate titles, labels, and units.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to combine data and then represent that data on a line plot.

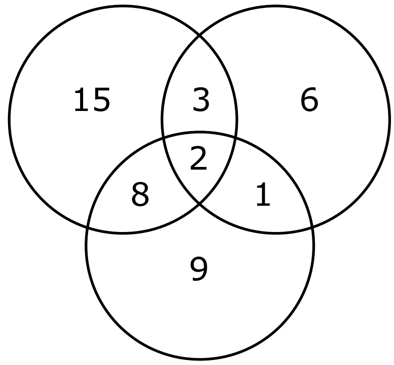
Distractor Rationale:

A. The student saw that the line plot was correct for red, blue, and yellow.
B. The student did not combine the girls and saw that the line plot was correct for Fran.
C. The student did not combine the girls and saw that the line plot was correct for Joy.
D. Correct. The student demonstrated an ability to represent data on a line plot marked with whole numbers.

28 **Jason asked his friends if they listen to certain types of music and recorded the information in this Venn diagram.**

Types of Music Listened To

Rock Jazz



Country

Which frequency chart displays the same data?

A **Types of Music Listened To**

Type	Frequency
Rock	15
Jazz	6
Country	9

B **Types of Music Listened To**

Type	Frequency
Rock	17
Jazz	8
Country	11

C **Types of Music Listened To**

Type	Frequency
Rock	26
Jazz	10
Country	18

D **Types of Music Listened To**

Type	Frequency
Rock	28
Jazz	12
Country	20

28 continued...

Standard: 4.D.1.2 Organize data sets to create tables, bar graphs, timelines, and Venn diagrams. The data may include benchmark fractions or decimals ($\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, 0.25, 0.50, 0.75).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate information from a Venn diagram to a table.

Distractor Rationale:

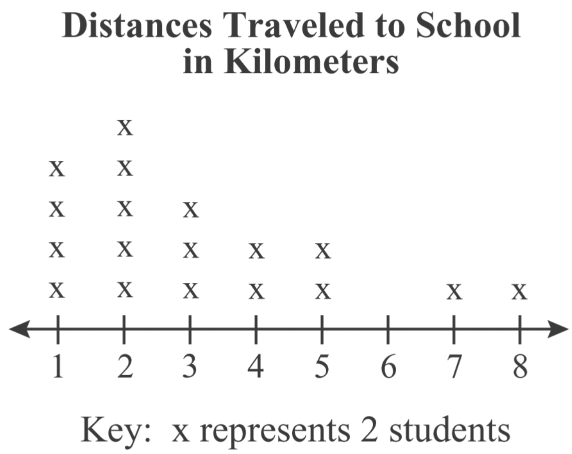
A. The student only used the numbers in the outermost parts of the Venn diagram, not taking into account the numbers shared by the different types of music.

B. The student only used the numbers in the outermost parts of the Venn diagram and the number in the very middle.

C. The student failed to include the 2 which is shared by all music types.

D. Correct. The student demonstrated an ability to translate data presented in a Venn diagram to a table.

29



What is the total number of students who are represented by this line plot?

A 34

B 36

C 54

D 56

Standard: 4.D.1.3 Solve one- and two-step problems by analyzing data in whole number, decimal, or fraction form in a frequency table and line plot.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine the total number of students represented on a line plot when the key represents more than 1.

Distractor Rationale:

A. The student missed one x.

B. Correct. The student demonstrated an ability to understand data presented on a line plot.

C. The student thought x represented 3 students.

D. Balance distractor

Cluster Items

The following sample items are part of a cluster. The cluster is presented first and then the two items that follow require use of the cluster. The two items are from different standards.

Use this information to answer the following questions.

At the beginning of the week, Gabriela had $12 and Henry had $9. During the week, they both earned money collecting cans that they recycled. At the end of the week, Gabriela and Henry each had $20.

30 **Gabriela took the money she earned to the movie theater. She bought a ticket and a drink for a total of $14. How much money did she have left?**

A $2

B $6

C $14

D $34

Standard: 4.N.4.2 Given a total cost (dollars and coins up to twenty dollars) and amount paid (dollars and coins up to twenty dollars), find the change required in a variety of ways.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the amount of money left after buying items.

Distractor Rationale

A. The student computed 14−12 instead of 20−12.

B. Correct. The student demonstrated an ability to find the amount of money left after paying.

C. The student gave the total spent instead of the amount left.

D. The student added instead of subtracted.

31 **The equation shown can be used to find out how much Henry earned during the week collecting cans that he recycled. The value of the $\square$ is the amount Henry earned.**

$$12 + 8 = 9 + \square$$

Which value can be placed in the $\square$ to make this equation true?

A 3

B 11

C 20

D 29

Standard: 4.A.2.3 Determine the unknown addend or factor in equivalent and non-equivalent expressions (e.g., $5 + 6 = 4 + \square$, $3 \cdot 8 < 3 \cdot \square$).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to decide how to determine the unknown addend in an equation.

Distractor Rationale

A. The student confused 12 and 20.

B. Correct. The student demonstrated an ability to determine the unknown addend in equivalent expressions.

C. The student knew the total must be 20, but failed to subtract the 9.

D. The student added 9 to 20 instead of subtracting.